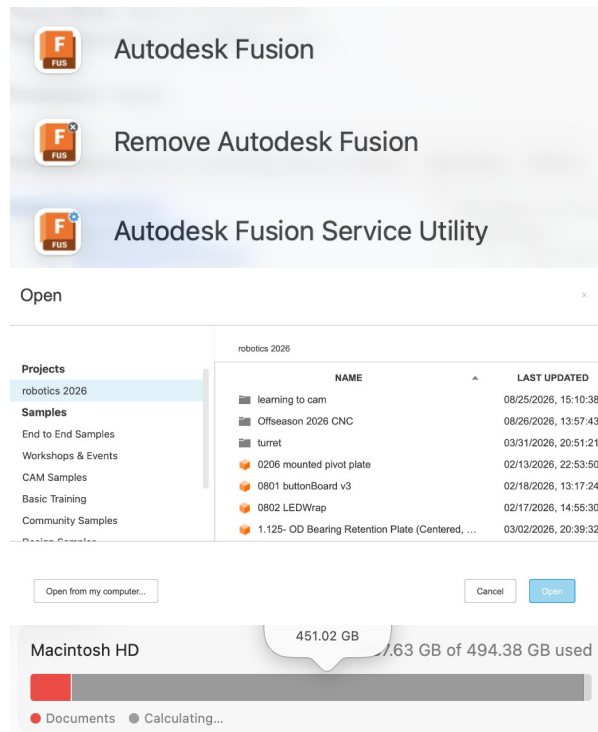


Name: Arnav Krishnan
Report Date: Week of August 24
Work Completion Conflicts: None

Completed Tasks

Downloading and Learning basic Fusion - Monday - 50min



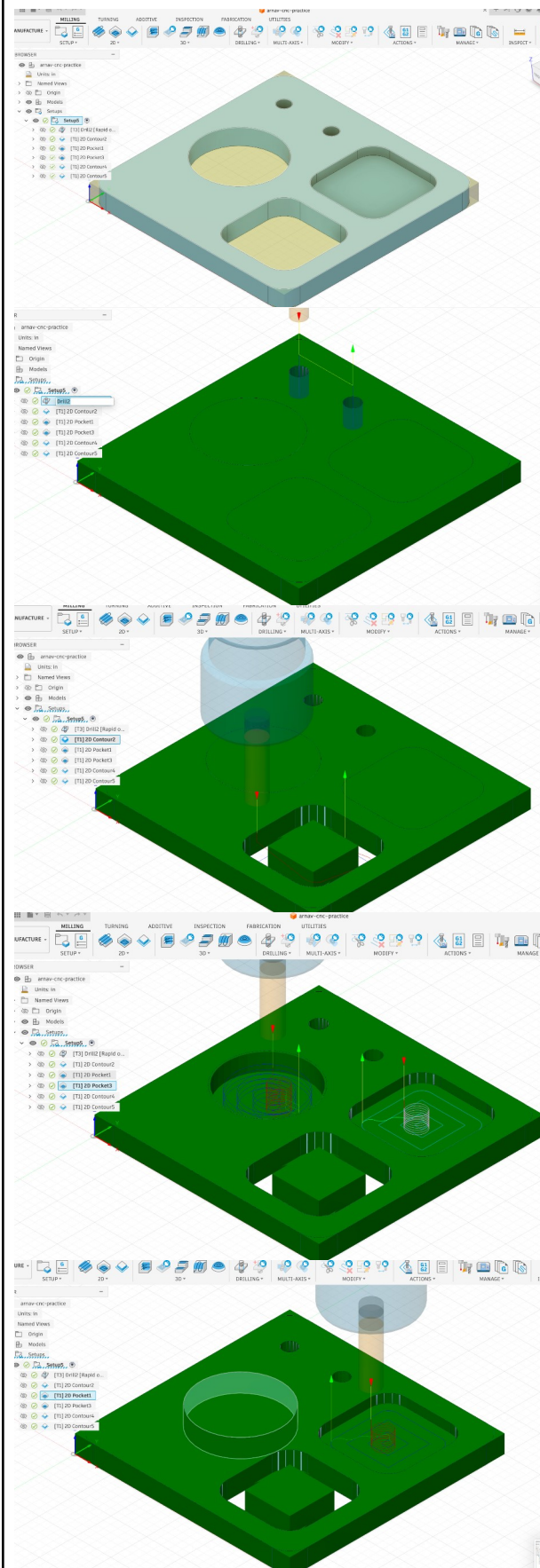
Fusion used up 20GB of storage... this is an issue :0 (above image)

I used our first class to set up a lot of the software that the rest of our classes that week would depend upon. The main one was Autodesk Fusion, which you can see the installations for in the evidence; Fusion was a very large download so it took a long time. This mattered a lot because our later CAD and CNC work depended on “existing” within a project structure, and one that is controlled and stays the same. For that reason, we choose to use Fusion for all our files.

In terms of individual contribution, mine was simply the initial software download and setup, and beginning to learn Fusion’s interface. I learned how the project browser, modeling workspace, and other tools fit into the workflow that one can use in Fusion. It was very cool to see how a single program supported sketching, solid features, simulation, CAD, machine output, etc., and I initially definitely felt a little bit in-over-my-head with all the features available.

The main “challenge” today wasn’t really anything particularly difficult, but rather just a practical cost of getting started. My mac was already running low on storage so I had to clear out a ton of old files, as Fusion used up 20 gigabytes of storage. This reduced my available disk space significantly. In the future I would check available storage before installing, remove unused installers or redundant applications, and confirm that the project files are stored in an organized location before beginning design work.

Designing our first pieces to be cut on the CNC in Fusion - Wednesday - 1hr 30min



During the design session, on Wednesday, I moved from learning the interface of Fusion to creating a part that could actually be manufactured. I modeled this part while working with Farhan.

Specifically, while there were elements of Fusion that I learned as I followed along with Farhan's work, the majority of basic learning I had already done yesterday, which made it much smoother for me today!

The model that we were designing, and would eventually create using the CNC, is a flat, rectangular plate with softened outside corners. Its top face contains several different feature types:

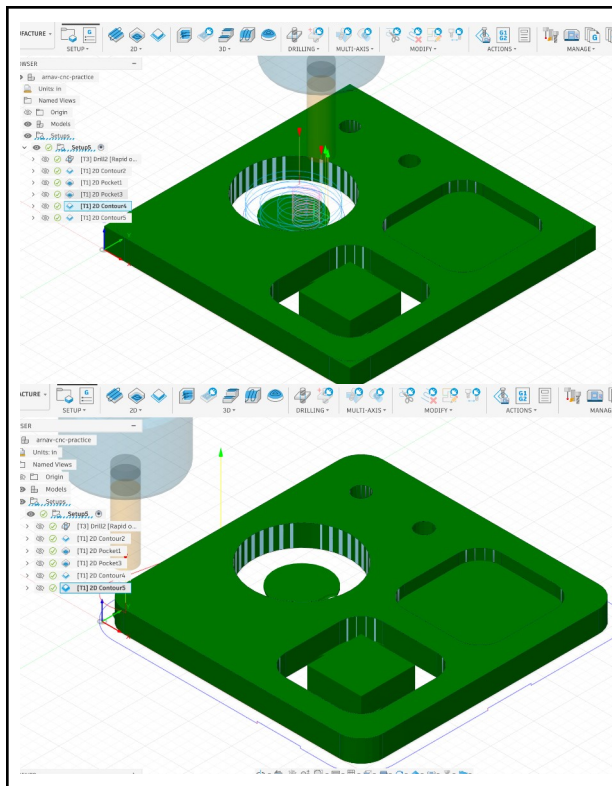
- a large circular opening
- two small circular holes
- a rounded-rectangle opening
- and a rounded-rectangle "pocket", in the sense that it does not go all the way through the piece

Designing several feature types in one part gave me a useful first test of my abilities, and forced me to learn a significant amount, because it required more than a single outline cut.

Fusion's manufacturing workspace was new to me, and how the tool would move through the stock was an interesting and useful feature, but one I had to learn as I went along. The

screenshots show the process becoming progressively more specific, and I tried to attach them in generally chronological order: first the solid model, then green stock and machine context, then visible contours and pocketing paths.

The Operation uses a number 3 drill with a 0.191 inch diameter and a 135 degree point, a spindle speed of approximately 5,999.56 rpm.



For me, the challenge was translating a visually simple plate into multiple machining decisions. A large opening and a rounded pocket, even if the same general square shape, have very different machining processes and thus have very different design processes.

DRILL : DRILL2



▼ Tool

Tool

Select...Edit...

#3 - Ø0.191" 135° drill (N11) ▼

▼ Feed & Speed

Preset

Default Pr... ▼

Spindle Speed

5999.560 rpm

Surface Speed

300.00 ft/min

Use Feed per Rev...

☐

Plunge Feedrate

18.00 in/min

Plunge Feed per F...

0.003 in

Retract Feedrate

50.00 in/min

Retract Feed per I...

0.008 in

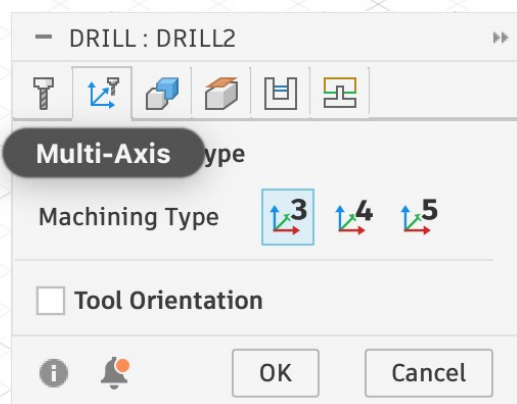
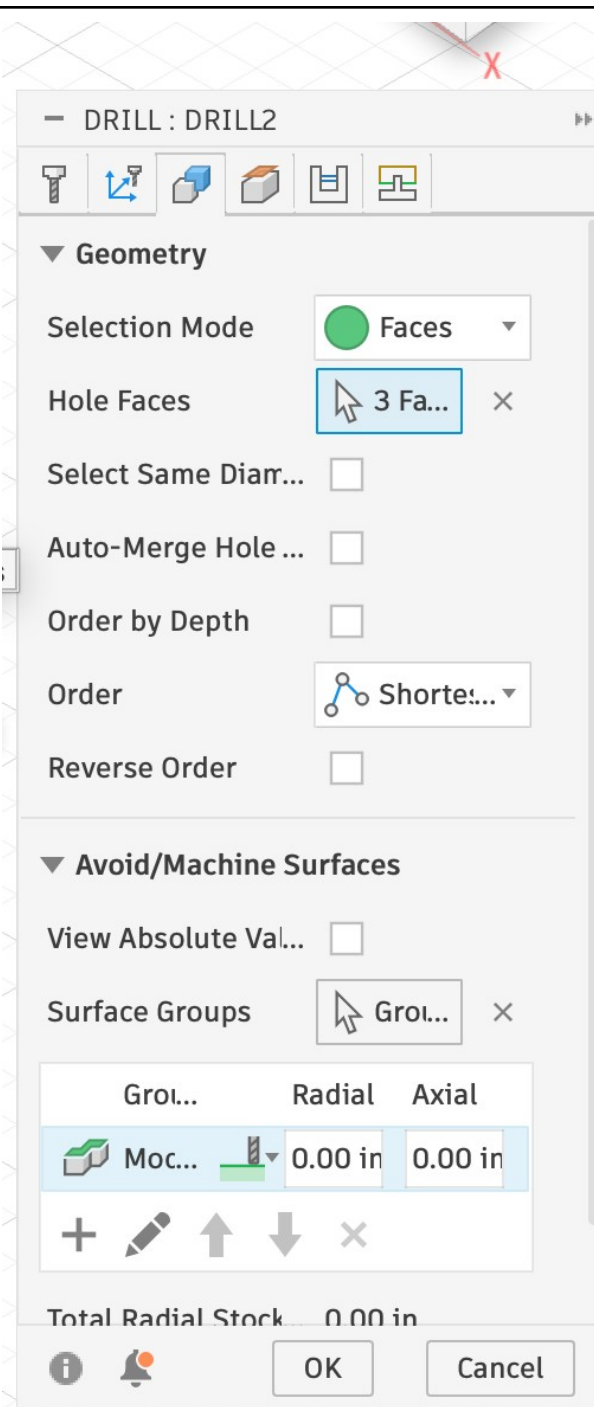
Coolant

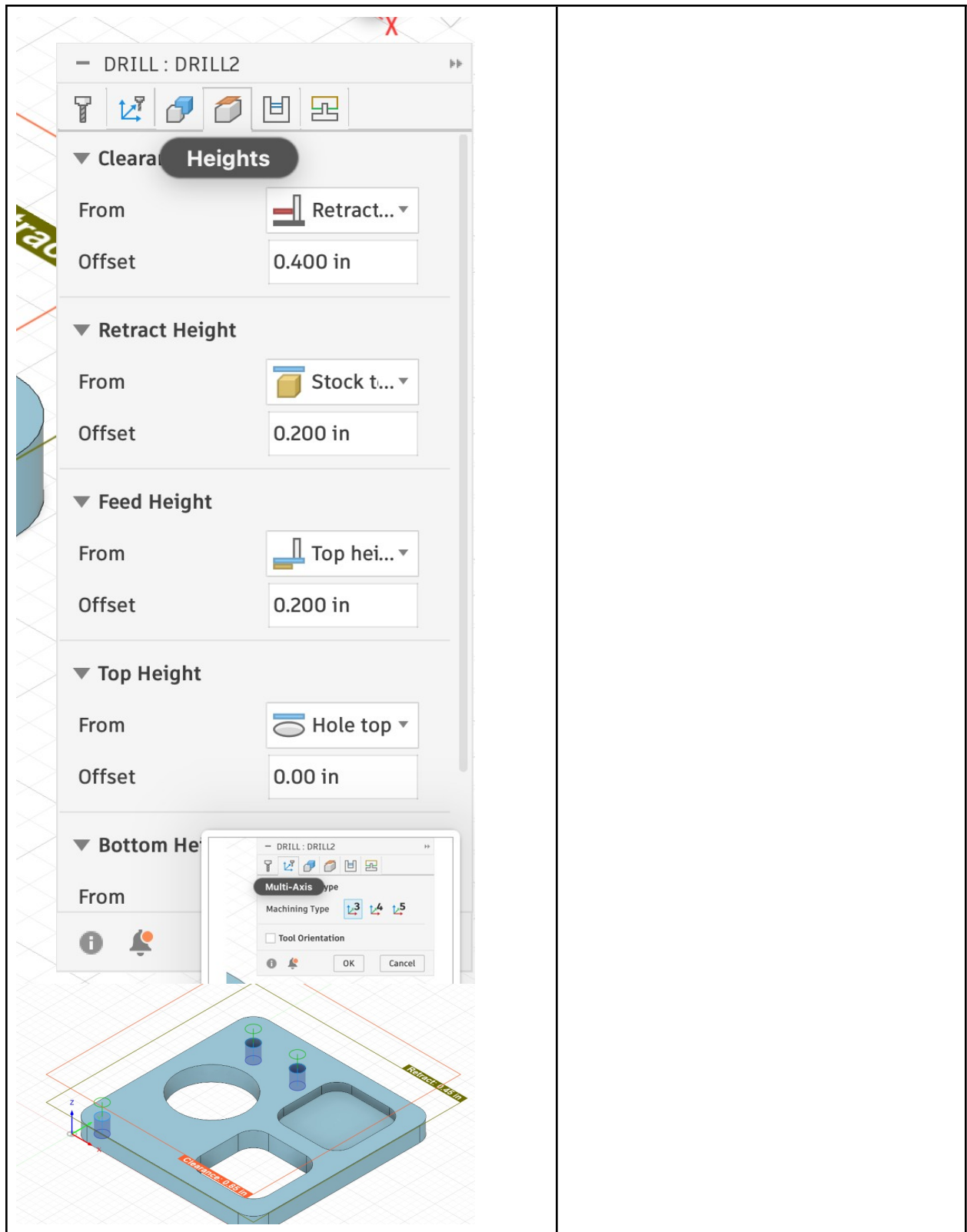
 Flood ▼

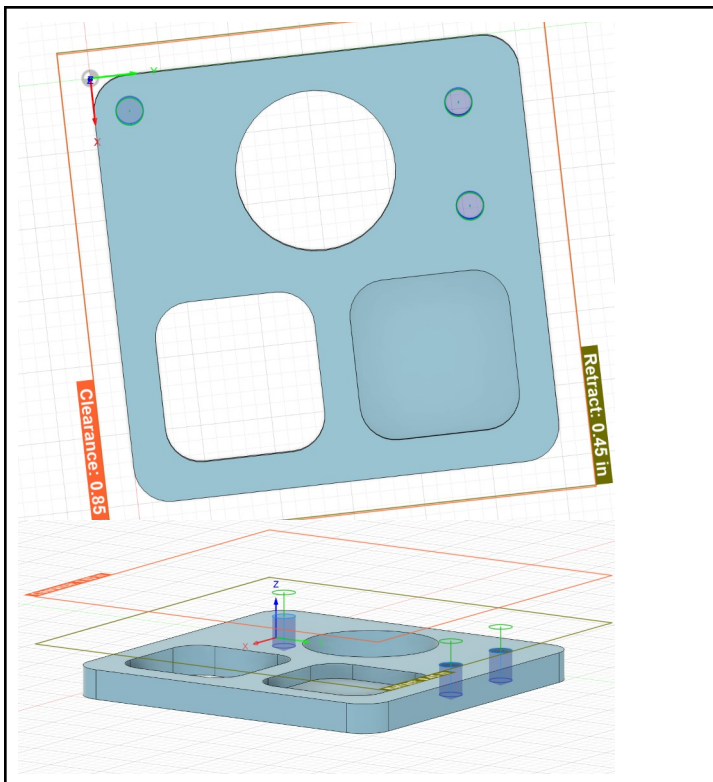
☐ Flute, Shaft & Holder



OKCancel







Using the CNC to cut the pieces - Friday - 1hr 30min



On Friday Farhan, I, and the rest of our group connected the prepared digital workflow to the physical CNC router. The photographs show the stock positioned on the router bed, secured with clamps in a few places (which is very important!), and cut while the operator, or in this case, all 4 operators, remained close to the piece while being cut. Farhan explained that while technically the CNC shouldn't need supervision, things will go wrong, things will break, and so it is very important to be there.

We all took turns operating the vacuum, and the bed contains chips from material removal. These details are important because the CNC process can be a bit sensitive in the sense that the material has to stay fixed, the cutting area has to remain observable, and chips have to be managed, otherwise they might interfere with the process, which can lead to bad results.

The finished transparent part broadly matches the Fusion model: it has:



- a large circular opening
- two small circular holes
- a rounded-rectangle opening
- and a rounded-rectangle “pocket”, in the sense that it does not go all the way through the piece

So, I would say that the CNC'ed piece was the main success of the task. At the same time, when you look at the close-up evidence, you may notice rough or chipped-looking material around parts of the large circular opening and visible marks on the surface, even after I deburred it.

Several variables could explain the imperfect edge quality, although we cannot be sure. Some possible factors include workholding vibration, which I personally believe is the most likely. In the future, I will make an attempt to control these to make better parts, for example by ensuring that the clamps are placed nearby the cutting area, and clamped very tight.

